

Brac University

CSE260: Digital Logic Design | Quiz – 1 | Full Marks – 3 + 6 + 6 = 15 | Time – 30 minutes | Summer 2025

Name:

Student ID:

Section:

1.CO1	Represent -67 with -75 in 11 bits using 2's complement system.
2.CO1	A construction project has three phases. Phase 1 costs $(A3)_{16}$ dollars, while Phase 2 costs $(124.2)_8$ dollars. Phase 3 has been operating for $(110)_5$ days, each day costs $(5.43)_6$ dollars. How much more money will be needed for Phase 3 to match the total cost of Phases 1 and 2 combined? (Show the answer in decimal, show the conversions)
3.CO1	Divide $(1112311)_4$ by $(123)_4$ using base 4. Find the quotient and remainder. Note: You must show the necessary calculations.

Solution

- $67 = 1000011$
 $+67 = 01000011$
 $+67$ in 11 bits = 00001000011
 -67 in 11 bits = 11110111100+1 = **11110111101**

$75 = 1001011$
 $+75 = 01001011$
 $+75$ in 11 bits = 00001001011
 -75 in 11 bits = 11110110100+1 = **11110110101**

- Phase 1 cost = $(A3)_{16} = 163$ dollars
 Phase 2 cost = $(124.2)_8 = 84.25$ dollars
 Phase 3 has been operating for = $(110)_5 = 30$ days
 Each day costs = $(5.43)_6 = 5.75$ dollars
 So, for 30 days = $30 \times 5.75 = 172.5$ dollars
 Total cost of Phase 1 and 2 = $163 + 84.25 = 247.25$
 Total money needed for Phase 3 to match the total cost of Phase 1 and 2 = $247.25 - 172.5 = \mathbf{74.75}$.

- $123 \overline{) 1112311} \quad \mathbf{3031}$
 $\phantom{123 \overline{) 1112311}} 1101$

$\phantom{123 \overline{) 1112311}} \underline{113}$
 $\phantom{123 \overline{) 1112311}} 0$

$\phantom{123 \overline{) 1112311}} \underline{1131}$
 $\phantom{123 \overline{) 1112311}} 1101$

$\phantom{123 \overline{) 1112311}} \underline{301}$
 $\phantom{123 \overline{) 1112311}} 123$

$\phantom{123 \overline{) 1112311}} \underline{\mathbf{112}}$

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1.CO1	Represent -59 with -83 in 11 bits using 2's complement system.
2.CO1	A construction project has three phases. Phase 1 costs $(A3)_{16}$ dollars, while Phase 2 costs $(124.2)_8$ dollars. Phase 3 has been operating for $(110)_5$ days, each day costs $(5.43)_6$ dollars. How much more money will be needed for Phase 3 to match the total cost of Phases 1 and 2 combined? (Show the answer in decimal, show the conversions)
3.CO1	Divide $(1112311)_4$ by $(123)_4$ using base 4. Find the quotient and remainder. Note: You must show the necessary calculations.

Solution

1. $59 = 111011$
 $+59 = 0111011$
 $+59$ in 11 bits = 00000111011
 -59 in 11 bits = 11111000100+1 = **11111000101**

$83 = 1010011$
 $+83 = 01010011$
 $+83$ in 11 bits = 00001010011
 -83 in 11 bits = 11110101100+1 = **11110101101**

2. Phase 1 cost = $(A3)_{16} = 163$ dollars
Phase 2 cost = $(124.2)_8 = 84.25$ dollars
Phase 3 has been operating for = $(110)_5 = 30$ days
Each day costs = $(5.43)_6 = 5.75$ dollars
So, for 30 days = $30 \times 5.75 = 172.5$ dollars
Total cost of Phase 1 and 2 = $163 + 84.25 = 247.25$
Total money needed for Phase 3 to match the total cost of Phase 1 and 2 = $247.25 - 172.5 = \mathbf{74.75}$.

3. $123 \overline{) 1112311}$ [**3031**
1101

$\begin{array}{r} 113 \\ 0 \end{array}$

$\begin{array}{r} 1131 \\ 1101 \end{array}$

$\begin{array}{r} 301 \\ 123 \end{array}$

$\begin{array}{r} 112 \end{array}$

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1.CO1	Represent -53 and -89 in 11 bits using 2's complement system.
2.CO1	A construction project has three phases. Phase 1 costs $(A3)_{16}$ dollars, while Phase 2 costs $(124.2)_8$ dollars. Phase 3 has been operating for $(110)_5$ days, each day costs $(5.43)_6$ dollars. How much more money will be needed for Phase 3 to match the total cost of Phases 1 and 2 combined? (Show the answer in decimal, show the conversions)
3.CO1	Divide $(1112311)_6$ by $(41)_6$ using base 6. Find the quotient and remainder. Note: You must show the necessary calculations.

Solution

- $53 = 110101$
 $+53 = 0110101$
 $+53$ in 11 bits = 00000110101
 -53 in 11 bits = 11111001010+1 = **11111001011**

$$\begin{aligned}
 89 &= 1011001 \\
 +89 &= 01011001 \\
 +89 \text{ in 11 bits} &= 00001011001 \\
 -89 \text{ in 11 bits} &= 11110100110+1 = \mathbf{11110100111}
 \end{aligned}$$

- Phase 1 cost = $(A3)_{16} = 163$ dollars
Phase 2 cost = $(124.2)_8 = 84.25$ dollars
Phase 3 has been operating for = $(110)_5 = 30$ days
Each day costs = $(5.43)_6 = 5.75$ dollars
So, for 30 days = $30 \times 5.75 = 172.5$ dollars
Total cost of Phase 1 and 2 = $163 + 84.25 = 247.25$
Total money needed for Phase 3 to match the total cost of Phase 1 and 2 = $247.25 - 172.5 = \mathbf{74.75}$.

- $41 \mid 1112311 \quad \mathbf{14231}$
41

$$\begin{array}{r}
 \hline
 302 \\
 244 \\
 \hline
 143 \\
 122 \\
 \hline
 211 \\
 203 \\
 \hline
 41 \\
 41 \\
 \hline
 0
 \end{array}$$

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Section:

1.CO1	Represent -47 and -95 in 11 bits using 2's complement system.
2.CO1	A construction project has three phases. Phase 1 costs $(A3)_{16}$ dollars, while Phase 2 costs $(124.2)_8$ dollars. Phase 3 has been operating for $(110)_5$ days, each day costs $(5.43)_6$ dollars. How much more money will be needed for Phase 3 to match the total cost of Phases 1 and 2 combined? (Show the answer in decimal, show the conversions)
3.CO1	Divide $(1112311)_6$ by $(41)_6$ using base 6. Find the quotient and remainder. Note: You must show the necessary calculations.

Solution

- $47 = 101111$
 $+47 = 0101111$
 $+47$ in 11 bits = 00000101111
 -47 in 11 bits = 11111010000 + 1 = **1 1 1 1 1 0 1 0 0 0 1**

$$\begin{aligned}
 95 &= 1011111 \\
 +95 &= 01011111 \\
 +95 \text{ in 11 bits} &= 00001011111 \\
 -95 \text{ in 11 bits} &= 11110100000 + 1 = \mathbf{1\ 1\ 1\ 1\ 0\ 1\ 0\ 0\ 0\ 0\ 1}
 \end{aligned}$$

- Phase 1 cost = $(A3)_{16} = 163$ dollars
Phase 2 cost = $(124.2)_8 = 84.25$ dollars
Phase 3 has been operating for = $(110)_5 = 30$ days
Each day costs = $(5.43)_6 = 5.75$ dollars
So, for 30 days = $30 \times 5.75 = 172.5$ dollars
Total cost of Phase 1 and 2 = $163 + 84.25 = 247.25$
Total money needed for Phase 3 to match the total cost of Phase 1 and 2 = $247.25 - 172.5 = \mathbf{74.75}$.

- 41] 1112311 [**14231**
41

$$\begin{array}{r}
 \hline
 302 \\
 244 \\
 \hline
 143 \\
 122 \\
 \hline
 211 \\
 203 \\
 \hline
 41 \\
 41 \\
 \hline
 0
 \end{array}$$